3.2.1

Wednesday, April 8, 2026 10:25 AM

Exercise 1. What is the absolute value of the integers -123, 27 and 0?

Exercise 2. Compute the factorization of 30030 and double-check your results using Sage.

Exercise 3. Consider the following equation:

$$4 \cdot x + 21 = 5$$

Compute the set of all solutions for x under the following alternative assumptions:

1. The equation is defined over the set of natural numbers.
2. The equation is defined over the set of integers.

Exercise 4. Consider the following equation:

$$2x^3 - x^2 - 2x = -1.$$

Compute the set of all solutions x under the following assumptions:

1. The equation is defined over the set of natural numbers.
2. The equation is defined over the set of integers.
3. The equation is defined over the set of rational numbers.

1. $|-123| = 123$
   $|27| = 27$
   $|0| = 0$
2. $30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$

$2 \quad 15015$

$3 \quad 5005$

$5 \quad 1001$

$7 \quad 143$

$11 \quad 13$

```
sage: n = NN(30030)
sage: factor(n)
2 * 3 * 5 * 7 * 11 * 13
```

3. $4x + 21 = 5$
   $4x = 5 - 21$
   $4x = -16$
   $x = -\frac{16}{4}$
   $x = -4$

1) $\emptyset$

2) $\{-4\}$

4. $2x^3 - x^2 - 2x = -1$
   $2x^3 - x^2 - 2x + 1 = 0$
   $x^2(2x-1) - 1(2x-1) = 0$
   $(2x-1)(x^2-1) = 0$
   $(2x-1)(x+1)(x-1) = 0$

   $2x - 1 = 0$ $\quad x + 1 = 0$ $\quad x - 1 = 0$
   $2x = 1$ $\quad x = -1$ $\quad x = 1$
   $x = \frac{1}{2}$

1) $\{1\}$

2) $\{-1, 1\}$

3) $\{-1, \frac{1}{2}, 1\}$